MGT 403 · PROBABILITY MODELING & STATISTICS · YALE SOM · DATA PROJECT

Where should we open our next preschool?

Pre-analysis and appendix: the data, methods and checks behind our answer

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We look for counties where parents pay much more for center-based preschool than local costs justify, where this advantage has lasted for years, and where enough families need care. This document answers the pre-analysis tasks and shows how each of our ideas became a measurable step, which data and samples each step uses, and how we tested whether the result can be trusted.

Our answer: ten counties

RANK	COUNTY	METRO AREA	AVERAGE PREMIUM 2015–22	EXPECTED PREMIUM	CHILDREN NEEDING CARE	SCORE
1	Arlington, VA	Northern Virginia	+61%	**+37%**	11,577	0.83
2	Alexandria, VA	Northern Virginia	+50%	**+31%**	8,813	0.77
3	San Francisco, CA	San Francisco	+36%	**+23%**	30,027	0.72
4	Loudoun, VA	Northern Virginia	+35%	**+22%**	22,329	0.70
5	Olmsted, MN	Rochester	+33%	**+21%**	9,533	0.60
6	Fairfax, VA	Northern Virginia	+30%	**+19%**	57,257	0.50
7	Multnomah, OR	Portland	+32%	**+20%**	32,227	0.45
8	Howard, MD	Baltimore	+32%	**+20%**	16,287	0.42
9	Hall, GA	Gainesville	+33%	**+21%**	9,215	0.40
10	Sullivan, TN	Kingsport	+28%	**+18%**	5,603	0.12

Premium = how much the county's preschool price exceeds the price predicted from its childcare-worker wage and rent, compared with counties in the same state (Appendix B). Expected premium = the average premium after the typical fading of premiums over about five years (Appendix D). Score = 50% premium rank + 50% demand rank among the ten (Appendix D).

How to read this document

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Team boxes contain the team's own interpretations and decisions (decisions were taken after the options and evidence were laid out; Appendix D). Data work, statistics and drafts of the text were produced with an AI tool (Claude Code, Python) at our direction; all prompts are documented in the prompt log submitted with the code. Beyond course scope marks methods that go beyond the course.

PART I

Pre-analysis

Project data work: prices and costs. Answers to the four pre-analysis tasks and the work behind them.

Task 1	The analysis dataset: merge, definition, sample flow	N = 8,146 county-years
Task 2	Descriptive statistics, 2022	N = 998 counties
Task 3	Prices vs. wages and rents: scatter plots and team interpretation	bivariate OLS
Task 4	Multivariate regression, model “c”: team decisions and interpretation	metro-clustered SEs

Pre-analysis: prices and costs (tasks 1–4)

OUR IDEA

Before searching for markets, establish how strongly childcare prices move with the two main costs of a center, staff wages and rent. The resulting model "c" is the starting point of our selection (Appendix B).

Task 1. The analysis dataset

Task. Merge NDCP county prices to the year × county cost/demand file from Problem Set 5, then define the analysis dataset as year × county observations where prices and predictor variables are non-missing.

- **Merge:** NDCP prices joined to the Problem Set 5 cost/demand panel on [year, 5-digit county FIPS], one-to-one. Both files use the same metro sample (OEWS area code starting with 00), so all 9,873 metro county-years matched.
- **Definition:** county-years with the center-based preschool price and all 10 predictors non-missing (wage, rent, per-capita income, population, children under 6, children under 6 whose resident parents all work, female college share, female full-time year-round share, under-6 share, care-need share). Requiring only price, wage and rent gives the identical sample in every year, so the definition does not depend on which predictors are counted. All regressions use this one sample.
- **Result: 8,146 county-years** of 1,159 counties (712 of them in all eight years); **2022: 998 counties** in 330 metro areas and 44 states including DC.

YEAR	METRO COUNTY-YEARS	NO NDCP PRICE	PRICE, NO WAGE	MISSING ACS	ANALYSIS DATASET	KEPT
2015	1,233	230	8	1	**994**	80.6%
2016	1,233	208	6	0	**1,019**	82.6%
2017	1,234	171	6	0	**1,057**	85.7%
2018	1,233	170	1	0	**1,062**	86.1%
2019	1,235	172	2	0	**1,061**	85.9%
2020	1,235	172	13	0	**1,050**	85.0%
2021	1,235	317	12	1	**905**	73.3%
2022	1,235	222	7	8	**998**	80.8%
Total	**9,873**	1,662	55	10	**8,146**	82.5%

Missing NDCP prices are the main reason for exclusion and are concentrated in a few states (2022: Puerto Rico, Indiana, Pennsylvania, Missouri, Arkansas). In 2021 NDCP has no prices for CA, NC, PA and VA. Missing ACS in 2022: the Connecticut counties (planning-region change).

Task 2. Descriptive statistics, 2022

Task. Make a table of mean / SD / number of observations for all variables in the analysis dataset for 2022.

VARIABLE (2022, METRO COUNTIES, 2022 $)	MEAN	SD	N	SOURCE
Outcome: childcare price				
Preschool price, center-based median ($/week)	174.48	59.75	998	NDCP MCPRESCHOOL
Infant price, center-based median ($/week)	214.03	82.62	997	NDCP MCINFANT
Toddler price, center-based median ($/week)	185.47	65.47	996	NDCP MCTODDLER
Costs				
Childcare-worker mean annual wage ($)	27,780.47	4,350.79	998	OEWS SOC 39-9011
Median gross rent ($/month)	1,105.97	336.40	998	ACS B25064_001E
Demand				
Per-capita income ($/year)	37,859.09	9,457.05	998	ACS B19301_001E
Total population	255,803.00	551,110.38	998	ACS B01001_001E
Own children under 6	17,151.19	37,250.04	998	ACS B23008_002E
Under 6, all resident parents in labor force	11,537.89	24,539.44	998	ACS B23008_004E+010E+013E
Share of women 25+ with bachelor's or higher	0.3062	0.1106	998	ACS B15002
Share of women 16-64 full-time year-round	0.4462	0.0559	998	ACS B23022
Share of population under 6	0.0644	0.0114	998	kids_under6 / pop_total
Share of population under 6 needing care	0.0432	0.0090	998	kids_under6_allpar_lf / pop_total

Unweighted county statistics. Infant and toddler prices are shown for context (not part of the complete-case definition). The sample excludes counties without NDCP prices, mainly lower-cost areas (no Puerto Rico counties remain), so cost and income means are above those for all metro counties (Problem Set 5: wage $27,373, rent $1,052).

Task 3. Prices vs. wages and rents

Task. Make scatter plots of childcare prices against (a) childcare-worker wages and (b) rents, with the fitted line from a linear regression. How do you interpret these?

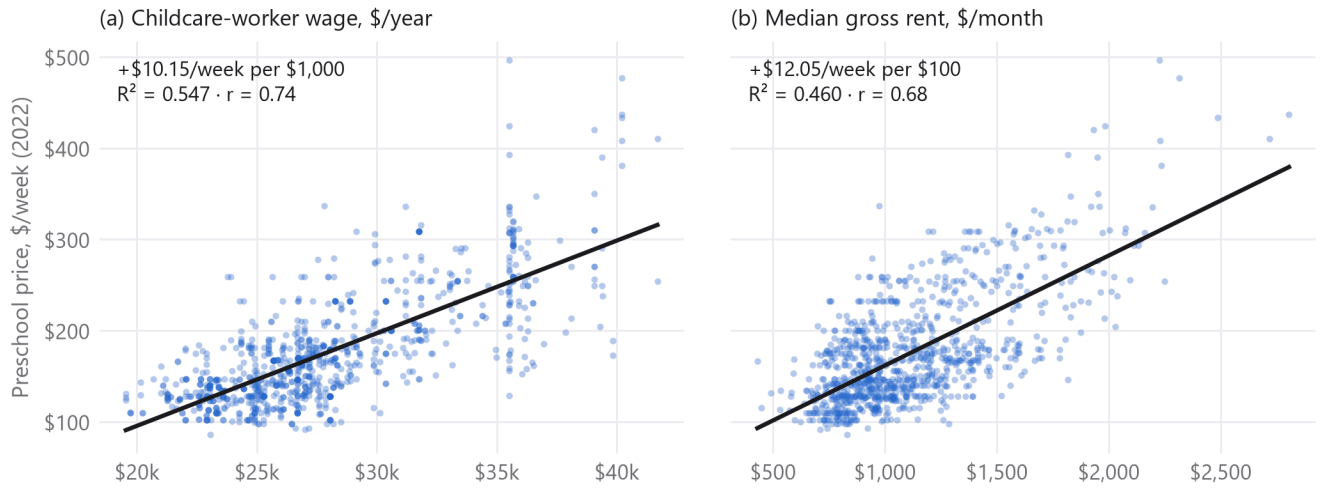


Figure P1. Center-based preschool price vs. costs, 2022, N = 998 metro counties; line = OLS fit. Vertical clusters in (a): all counties of a metro area share one OEWS wage but have different NDCP prices.

BIVARIATE OLS, 2022	FITTED LINE	SLOPE (READABLE UNITS)	R^2	CORRELATION
(a) Wage	−107.59 + 0.01015 × wage	+$10.15/week per $1,000	0.547	0.739
(b) Rent	41.17 + 0.1205 × rent	+$12.05/week per $100	0.460	0.679

TEAM INTERPRETATION · TASK 3

Both plots show a strong relationship between childcare prices and operating costs.

Childcare worker wage

Counties with higher childcare workers' wages tend to have higher weekly preschool prices. A $1,000 increase in annual worker wages corresponds with a $10.15 increase in weekly childcare prices. The R^2 indicates that the wages explain 54.7% of the variation in childcare prices.

Rent

Counties with higher median monthly rents tend to have higher childcare prices. A $100 increase in monthly rent is associated with an increase of about $12.05 in weekly childcare prices. The R^2 indicates that rent alone explains 46% of the variation in childcare prices.

Wages have a stronger relationship with childcare prices than rent, based on the respective R^2. The two plots show considerable variation around the line, meaning that even with similar costs the childcare prices can vary. That is especially visible in the vertical clusters around, for example, $35,000. The findings suggest both are important predictors of childcare prices, but there are other factors at play. Wages and rents are also correlated with each other (r = 0.67), since expensive metro areas tend to have both, so each line partly reflects the other cost.

Task 4. Multivariate regression

Task. Using 2022 data, regress median preschool price on (a) wages alone, (b) rent alone, and (c) wages and rent together. Interpret the coefficients in each, and explain any changes in the wage/rent coefficients between the univariate and multivariate specifications.

TEAM DECISIONS · TASK 4 (PARAPHRASED FROM TEAM NOTES)

- **Readable units:** price in $/week, wage per $1,000 of annual wage, rent per $100 of monthly rent (2022 $).
- **Standard errors clustered by metro area,** for the same reason as in Problem Set 5 (task 7): counties in one metro area share one OEWS wage, so they are not independent observations.

Model "c": $\text{Price}_i = a + b_1 \cdot \text{Wage}_i + b_2 \cdot \text{Rent}_i + e_i$

DEPENDENT VARIABLE: PRESCHOOL PRICE ($/WEEK)	(A) WAGE	(B) RENT	(C) WAGE + RENT
Wage (per $1,000/year)	**10.153** (0.647)	–	**7.093** (0.584)
Rent (per $100/month)	–	**12.053** (0.996)	**5.910** (0.922)
Intercept	−107.587 (17.103)	41.173 (9.797)	−87.946 (13.017)
R^2	0.547	0.460	0.608
Residual SD ($/week)	40.26	43.91	37.47
N (metro clusters)	998 (330)	998 (330)	998 (330)

Standard errors in parentheses, clustered by metro area. All slope coefficients p < 0.0001. The same 998 counties in all three models.

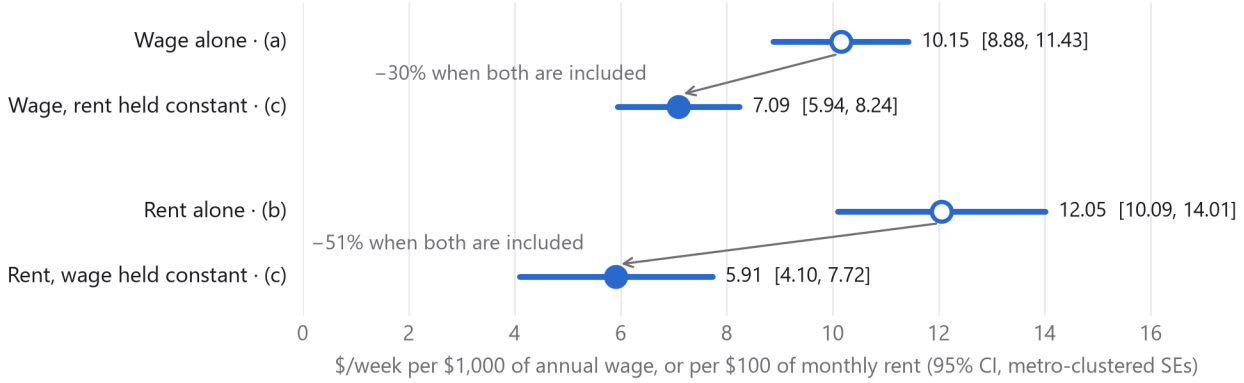


Figure P2. Cost coefficients alone (open) and together (filled), 2022, with 95% confidence intervals.

Omitted-variable decomposition (exact). Rent rises on average by 0.518 × $100 per $1,000 of wage, and wage by 0.866 × $1,000 per $100 of rent (r = 0.67). Hence $b_{wage}(a) = 7.093 + 5.910 \times 0.518 =$ **10.153** and $b_{rent}(b) = 5.910 + 7.093 \times 0.866 =$ **12.053**, exactly the single-regression coefficients. Holding rent constant, moving a county from the 25th to the 75th percentile of wages raises the predicted price by $35.82/week; the same move in rent adds $23.92/week. The intercepts are not interpretable on their own (a wage or rent of $0 is far outside the data).

ROBUSTNESS: SE OF THE SLOPE	OLS	METRO-CLUSTERED (USED)	STATE-CLUSTERED	P, STATE-CLUSTERED
(a) wage alone	0.293	**0.647**	0.660	4.4e-19
(b) rent alone	0.413	**0.996**	0.936	2.4e-16
(c) wage, rent held constant	0.367	**0.584**	1.052	3.0e-08
(c) rent, wage held constant	0.475	**0.922**	1.059	1.5e-06

TEAM INTERPRETATION · TASK 4

The regressions show that both childcare worker wages and rent have a positive and significant relationship with weekly preschool prices.

In “a”, a $1,000 increase in annual worker wages corresponds with a $10.15 increase in weekly childcare prices.

In “b”, a $100 increase in monthly rent is associated with an increase of about $12.05 in weekly childcare prices.

In “c”, when wages and rent are both included, their coefficients become smaller compared to their individual coefficients. Holding rent constant, a $1,000 increase in annual wages is associated with a $7.09 increase in weekly preschool prices. Similarly, holding wages constant, a $100 increase in monthly rent is associated with a $5.91 increase in weekly prices.

Both variables are positively correlated with r = 0.67, meaning that counties with higher wages also tend to have higher rents. Because wages and rent are positively correlated, the coefficient in each single regression also captures part of the effect of the other, omitted cost (omitted-variable bias). Once both are included, each coefficient only measures its own effect, so both become smaller. The R^2 also increases to 0.608 in model “c”, meaning that wages and rent together explain about 61% of the variation in childcare prices. This is an improvement over either model alone, although approximately 39% of the variation remains unexplained.

PART II

Appendix

Final project analysis: data handling, methods, decisions and checks behind our ten recommended counties.

0	From our ideas to measurable steps; which years enter which analysis	
A	Data sources, variables, samples and data-handling decisions	all slides
B	Cost model: from model “c” to our prediction; fit and residuals	slide 2
C	Out-of-sample validation of the cost model	slide 2
D	Team decisions, selection rule, candidates, score; stability	slides 3, 5
E	Reliability: backtest, robustness, portfolio view, statistical checks, data quality	slide 4
F	Risk checks, public pre-K, the Northern Virginia cluster, other candidates	slide 6
G	Limitations and reproducibility	

APPENDIX 0

From our ideas to measurable steps

Our selection follows six guiding ideas the team set at the start of the project. The table shows how each idea was turned into a measurable step, which data it uses, and where the evidence is.

OUR IDEA	HOW WE MEASURED IT	DATA	APP.
Find counties where prices are unusually high relative to costs (wages and rent).	Cost model: predict each county's price from its childcare-worker wage and rent, within its state; premium = actual vs. predicted price.	NDCP prices, BLS OEWS wages, ACS rents, 2015–2022	B, C
The advantage must have lasted over time.	Rank on the average premium over all observed years (≥ 5); require a positive premium in every year; measure how much of a premium survives.	Yearly premiums 2015–2022; direct NDCP survey years	D
Prioritize places with strong childcare demand.	Market-size filter (≥ 4,286 children under 6 whose parents all work) and a demand score (children needing care, their share of the population, income).	ACS 5-year 2022	D
Start with a top 20, then add factors to get to the top 10.	Candidate pool = 20 largest average premiums; then persistence, market size and the score.	Selection rule, Appendix D	D
Test the approach with historical data to build the best predictive model.	Out-of-sample validation of the cost model; backtest of the complete rule as if decided in 2018; robustness to each design choice.	2015–2018 → 2022; 2015–2020 → 2022	C, E
Look beyond the model before investing.	Five local risk checks: competition, child-population trend, wage and rent growth, free public pre-K.	Census CBP 2022; ACS; OEWS; state/county pre-K rules	F

Which years enter which analysis

The rule never uses information from after the year it is tested on: the backtest decides with 2015–2018 data only and is judged on 2022; the validation of the cost model fits 2015–2020 and predicts 2022.

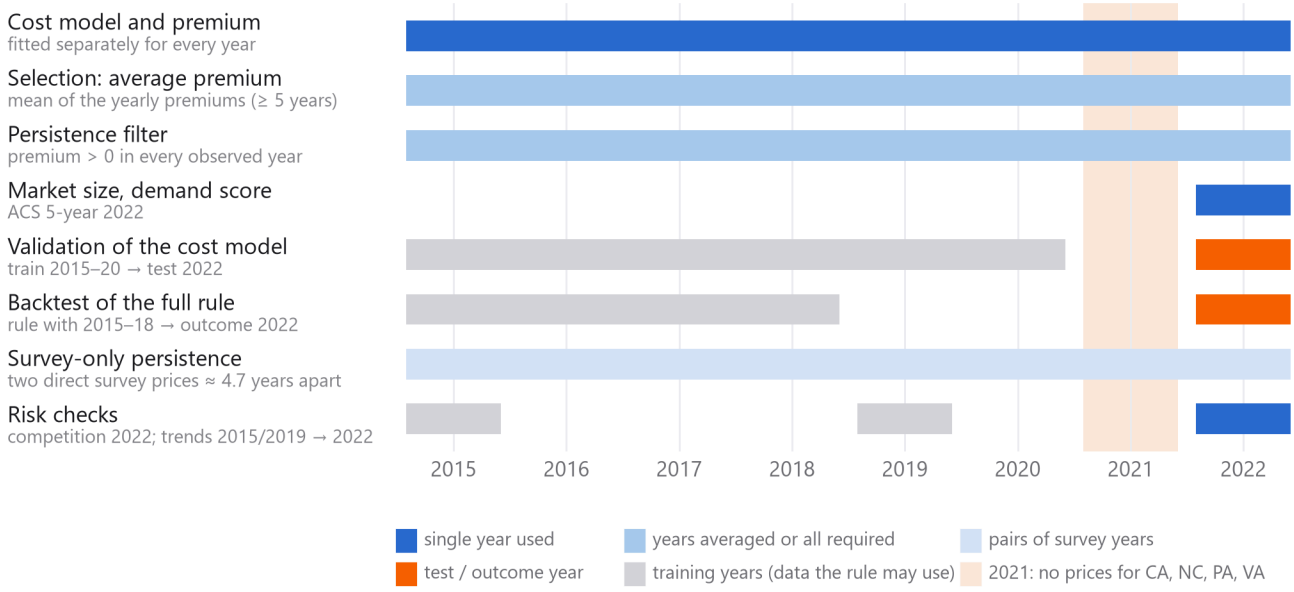


Figure 0.1. Years used by each analysis. Market size and demand use the 2022 ACS 5-year estimates (survey years 2018–2022).

APPENDIX A · ALL SLIDES

Data sources, samples and data handling

OUR IDEA

Use only official, county-level public data that a provider could also use, keep one consistent sample for every model, and document every choice that changes which observations enter.

A1. Sources

SOURCE	WHAT WE USE	GEOGRAPHY	YEARS
National Database of Childcare Prices (NDCP), U.S. Department of Labor	Median weekly full-time price of center-based preschool care (MCPRESCHOOL), imputation codes	County	2015–2022
Occupational Employment and Wage Statistics (OEWS), BLS	Mean annual wage of childcare workers (SOC 39-9011); preschool teachers (25-2011) for robustness	Metro area → its counties	2015–2022
American Community Survey 5-year (ACS), Census API	Median gross rent, children under 6, children whose resident parents all work, population, per-capita income	County	2015–2022
CPI-U (CUUR0000SA0), BLS	Annual average; converts all dollar values to 2022 dollars	U.S.	2015–2022
County–OEWS crosswalk, course package	Year-specific link of counties to OEWS areas; metro = area code starting with 00	County	2015–2022
County Business Patterns (CBP), Census API	Child day care establishments and employees (NAICS 624410)	County	2016, 2019, 2022
Gazetteer and cartographic boundaries, Census	County internal points, state and county outlines (maps only)	County	2022
Public pre-K program rules, state and county agencies	Eligibility and scope of free public pre-K (sources in Appendix F)	County / state	accessed Oct 2026

A2. Variables

VARIABLE	DEFINITION AND SOURCE CODE	UNIT	USED FOR
Price	NDCP MCPRESCHOOL × CPI factor	$/week, 2022 $	Cost model (dependent variable)
Childcare-worker wage	OEWS A_MEAN, SOC 39-9011, metro area	$/year, 2022 $	Cost model
Rent	ACS B25064_001E median gross rent	$/month, 2022 $	Cost model
Children needing care	ACS B23008_004E + 010E + 013E (under 6, all resident parents in the labor force)	children	Market-size filter, demand score
Care-need share	children needing care / ACS B01001_001E population	share	Demand score
Income per capita	ACS B19301_001E	$/year, 2022 $	Demand score
Premium (within state)	exp(residual) − 1 of the final cost model, per year	%	Selection, stability
National premium	same model without state effects	%	Robustness filter (> 0)
Direct survey year	5th digit of the NDCP imputation code = 0 (not interpolated)	flag	Survey-only persistence, data quality
Center density	CBP establishments per 1,000 children needing care	per 1,000	Risk check (competition)

A3. Samples

Analysis dataset (pre-analysis task 1): 8,146 metro county-years with the preschool price and all predictors non-missing; sample flow by year in Task 1.

Selection sample. Of the 1,159 counties with at least one year in the analysis dataset, 998 are in the 2022 data, 972 of these are observed in at least 5 of the 8 years, and 460 of those also have a positive average national premium. These are the counties eligible for the candidate pool (Appendix D). The 2022 cross-section has 998 counties in 330 metro areas and 44 states including DC.

A4. Data-handling decisions

ISSUE	WHAT WE DID	WHY	EFFECT
Metro sample	County-year is metro if its OEWS area code in the year-specific crosswalk starts with 00	Assignment definition; wages exist for metro areas	1,236 counties, 9,873 county-years
Wage geography	Every county gets the OEWS wage of its metro area	OEWS publishes no county wages	Counties of one metro share a wage → SEs clustered by metro area
Large metros 2015–2017	Use BLS whole-MSA files (aMSA); metropolitan divisions not used	Crosswalk uses whole-MSA codes	No wage gaps for the 11 largest metros
Legacy FIPS	Drop 12025 (Dade) and 51515 (Bedford city)	No longer counties; successors are in the crosswalk	Identical county set across sources
Connecticut 2022	No ACS values for the 8 old counties (ACS 2022 reports planning regions)	Geographies not comparable	CT drops out in 2022
Real dollars	All $ × CPI-U annual average factor to 2022	Compare years	Prices, wages, rents, income in 2022 $
Complete cases	Keep county-years with price and all predictors non-missing	One consistent sample for every model	8,146 county-years, 1,159 counties
NDCP gaps	2021: no prices for CA, NC, PA, VA; years kept as missing, not filled	Avoid inventing data	Those counties have 7 of 8 years
CA/VA 2022 prices	Kept, but selection uses the 2015–2022 average, not 2022 alone	NDCP extrapolated them (no 2022 survey); nominal change vs. 2020: SF +25%, Marin +37%, Los Angeles −15%, Arlington +21%, Fairfax −7%	Team decision D2
Candidates need a 2022 price	Pool drawn from counties in the 2022 sample	Current market evidence; demand measured in 2022	2 counties without 2022 price would otherwise enter (App. D)
Identifiers	FIPS, OEWS codes and imputation codes stored as text	Keep leading zeros	Exact joins
Raw data	Never edited; SHA-256 checksums of all downloaded files	Reproducibility	data/raw_checksums.sha256

APPENDIX B · SLIDE 2

The cost model: from model "c" to our prediction

OUR IDEA

If wages and rents explain what childcare *should* cost in a county, then a county where parents pay far more than that has pricing power: willingness to pay that local costs do not explain. That gap is the premium we look for.

B1. Starting point: model "c" from the pre-analysis

Price = −87.95 + 7.093·wage + 5.910·rent (2022, N = 998, price in $/week, wage per $1,000/year, rent per $100/month; SEs clustered by metro area: 0.584 and 0.922; R^2 0.608). Alone, wage has a slope of 10.153 and rent 12.053; both shrink together because wages and rents are correlated (omitted-variable bias, pre-analysis task 4). Model "c" already explains 61% of price differences, but three features of the data argue for refining it:

1. **States set price levels.** NDCP prices come from state market-rate surveys, often reported by rate zones: the 24 Wisconsin counties in 2022 have only 4 distinct prices. States also differ in staffing ratios and licensing rules that we do not observe. A national model therefore mostly finds "expensive states", not expensive markets.
2. **The scatter fans out.** Price differences grow with the price level, so a $50 gap means much more in Tennessee than in San Francisco. A model in logs measures the gap in percent and fits this pattern.
3. **Interpretation.** In logs the coefficients are elasticities: +10% wage → +3.6% price; +10% rent → +5.2% price.

B2. Four candidate models, 2022

MODEL	UNITS	WAGE	RENT	STATE EFFECTS	R^2	N
Model "c": level	$/week, levels	7.093 (0.584)	5.910 (0.922)	–	0.608	998
Log-log	elasticity	1.111 (0.072)	0.319 (0.038)	–	0.580	998
Level + state effects	$/week, levels	2.075 (0.539)	9.627 (1.124)	yes	0.801	998
Log-log + state effects (final)	**elasticity**	**0.374** (0.088)	**0.528** (0.043)	**yes**	**0.807**	**998**

SEs (in parentheses) clustered by metro area, because counties in one metro area share one OEWS wage. State effects: 43 dummies (44 states incl. DC). With state effects, the wage coefficient falls (2.08 vs. 7.09 in levels): part of the wage effect in model "c" was a state effect. Out-of-sample comparison of all four models: Appendix C.

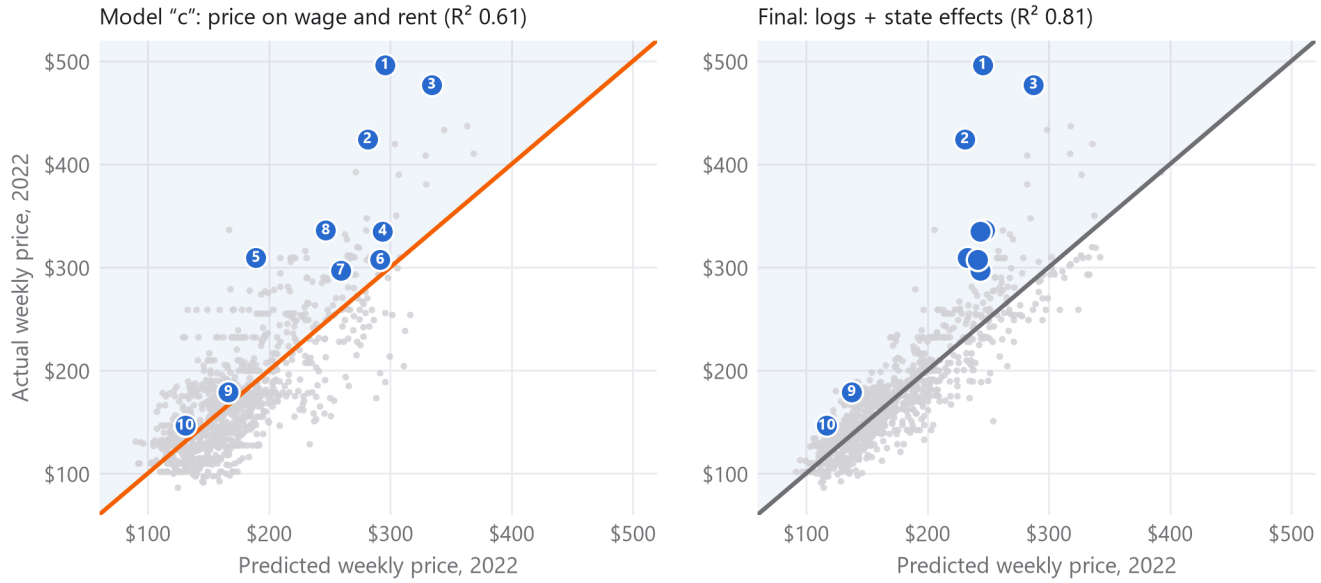


Figure B1. Actual vs. predicted weekly price, 2022 (numbers = our ranks). Above the line, parents pay more than costs justify. In model “c”, Hall (9) and Sullivan (10) look ordinary; compared within their own states, all ten lie clearly above the line.

TEAM DECISION D1 · PREMIUM MEASURE

Final model (estimated separately for each year 2015–2022): log(price) = a + b_1·log(wage) + b_2·log(rent) + state effects. **Premium** = exp(residual) − 1, i.e. the % by which the actual price exceeds the price predicted for a county with the same wage and rent *in the same state*. As a robustness requirement, the county's average national premium (same model without state effects) must also be positive, so that we do not pick counties that are merely pricey for a cheap state.

B3. Worked example: Hall County, GA (rank 9), 2022

Parents pay **$179/week**. Model “c” predicts $167 (+7%, an ordinary market). Compared within Georgia, the final model predicts **$138**, so Hall has a premium of **+30%**: Georgia's prices are low overall, and Hall is expensive for Georgia.

B4. Is the cost model stable over time?

The final model is fitted separately for each year, so each year's premium is measured against that year's cost relationship. The coefficients are stable: the wage elasticity stays between 0.28 and 0.37, the rent elasticity between 0.45 and 0.62, and R^2 between 0.81 and 0.89 (model "c": 0.56–0.67).

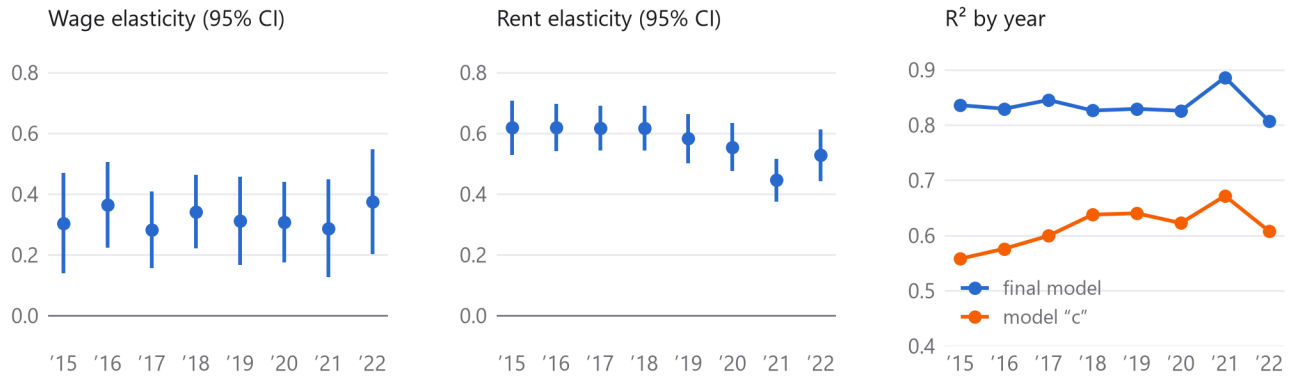


Figure B2. Final model by year: elasticities with 95% CIs (SEs clustered by metro area) and R^2 vs. model "c". 2021 has fewer observations (no NDCP prices for CA, NC, PA, VA).

YEAR	N	STATES	WAGE ELAST.	RENT ELAST.	R^2 FINAL	R^2 MODEL "C"	RESID. SD (LOG)
2015	994	43	0.30 (0.08)	0.62 (0.05)	0.84	0.56	0.121
2016	1,019	44	0.36 (0.07)	0.62 (0.04)	0.83	0.58	0.119
2017	1,057	46	0.28 (0.06)	0.62 (0.04)	0.85	0.60	0.117
2018	1,062	46	0.34 (0.06)	0.62 (0.04)	0.83	0.64	0.122
2019	1,061	46	0.31 (0.07)	0.58 (0.04)	0.83	0.64	0.120
2020	1,050	46	0.31 (0.07)	0.55 (0.04)	0.83	0.62	0.122
2021	905	42	0.29 (0.08)	0.45 (0.04)	0.89	0.67	0.114
2022	998	44	0.37 (0.09)	0.53 (0.04)	0.81	0.61	0.139

B5. Fit and residuals

81%
of 2022 price variation explained

±15%
typical miss (1 residual SD = 0.139 log points)

24
of 998 counties above the 95% prediction band (2.4%)

p < 0.001
Breusch–Pagan test: heteroskedasticity

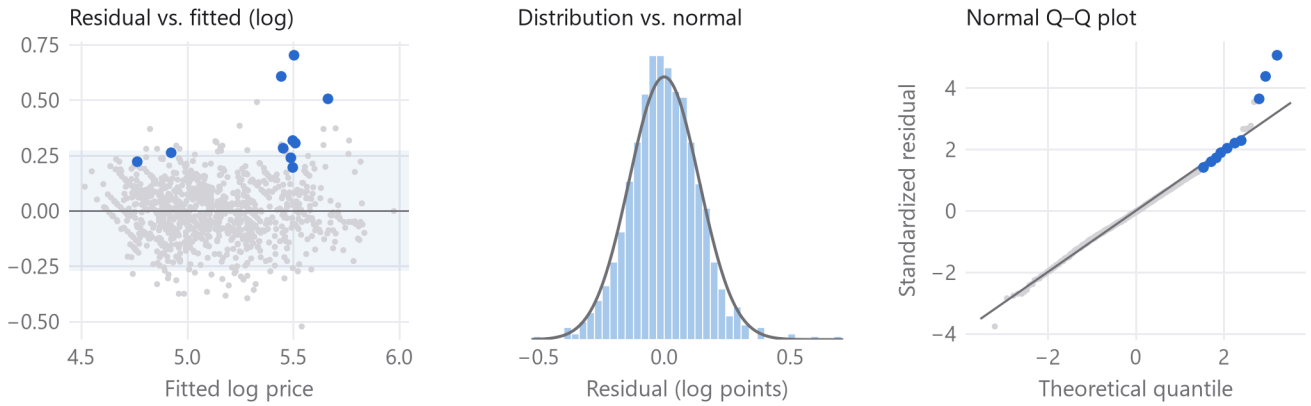


Figure B3. Residuals of the final model, 2022 (blue = top 10). Shaded band: ±1.96 residual SD (approximate; the count of counties above the 95% band uses the exact prediction interval).

- **No systematic pattern:** residuals are centered on zero across the range of fitted prices (correlation with fitted values = 0 by construction, and no visible curvature). Mean residuals are close to zero in every county-size quartile.
- **Shape:** close to normal (skewness 0.21, excess kurtosis 1.27); the Q–Q plot departs from the line only in the upper tail, which is exactly where our top 10 lie. These are real outliers in the sense that interests us, not a misfit of the bulk.
- **Heteroskedasticity:** the Breusch–Pagan test rejects constant variance (p = 6.1e-10). We therefore use standard errors clustered by metro area throughout, which are robust to unequal variance and to shared wages within a metro area.
- **What the residual contains:** 19% of price variation is not explained by wages, rents and states. It contains willingness to pay (our signal), but also quality differences, local regulation, unobserved costs and NDCP measurement error (our noise). Two features of our rule keep noise from driving the result: we average the premium over up to eight years, and we require it to be positive in every year (Appendix D).
- **Demand variables are deliberately not in the cost model.** Including income or children would explain part of the premium away; we use them afterwards to choose among counties with a premium.

APPENDIX C · SLIDE 2

Does the cost model predict prices it has not seen?

OUR IDEA

A model that only fits the data it was estimated on could be over-fitted. We want "the best predictive model possible", so we judged the candidates by how well they predict prices they have never seen.

Beyond course scope R^2 measures fit on the estimation sample. Out-of-sample tests hold data back, fit the model on the rest, and measure the typical error on the held-back prices (root mean squared error, RMSE, and mean absolute error, MAE, in $/week):

- **New year:** fit on 2015–2020 (pooled, with year effects), predict every county's 2022 price (N = 997).
- **New metro areas:** 2022, ten-fold cross-validation in which all counties of a metro area are always held out together, so the model never sees a county's neighbors or its shared wage (N = 992).

MODEL	R^2 (2015–20 FIT)	RMSE NEW YEAR	MAE NEW YEAR	RMSE NEW METROS	MAE NEW METROS
Model "c": level	0.606	$39.22	$30.85	$37.60	$28.42
Log-log	0.581	$39.08	$30.74	$37.35	$27.99
Level + state effects	0.810	$33.50	$26.03	$29.23	$20.86
Log-log + state effects (final)	**0.805**	**$33.67**	**$25.62**	**$30.38**	**$20.81**

−14%
forecast error for a new year vs. model "c"

−19%
error for unseen metro areas vs. model "c"

State effects
drive the gain; logs alone do not

LLS ≈ LS
similar accuracy; logs give % premiums

Reading. Adding state effects reduces the out-of-sample error by about a sixth in both tests; taking logs alone does not. The level and log models with state effects predict about equally well, so we chose the log version (D1) because it measures premiums in percent, comparable between cheap and expensive markets. Both errors refer to prices; the premium of a county is its prediction error by design, so a model that predicts well in general makes a large, persistent residual more meaningful.

APPENDIX D · SLIDES 3 AND 5

Team decisions, the selection rule, and stability over time

OUR IDEA

Start broad, then narrow down: take the 20 counties with the largest lasting price advantage, keep those whose advantage held every year and whose market is big enough, and rank the rest by price advantage and demand together.

D1. Team decisions

Each decision was taken by the team after the options and the evidence were laid out (the prompts are in the prompt log).

#	DECISION	ALTERNATIVES CONSIDERED	WHY
P1	Readable units: price $/week, wage per $1,000/year, rent per $100/month	Raw dollar units	Coefficients that non-statisticians can read (pre-analysis)
P2	Standard errors clustered by metro area	OLS, state clusters	Counties in one metro share one OEWS wage; not independent
D1	Within-state log premium (final model); national premium > 0 as robustness	Model "c" residual; national log model	State effects cut out-of-sample error by 14–19% (App. C); logs give % premiums, comparable across markets
D2	Rank on the average premium 2015–2022 (≥ 5 years), not on 2022 alone	2022 premium only	Goal is a lasting advantage; CA/VA 2022 prices are extrapolations (App. A)
D3	Premium positive in every observed year	One weak year allowed	The advantage must have lasted
D4	Market size ≥ median children needing care (4,286)	25th percentile; no filter	A high premium in a tiny market cannot fill a center
D5	Score = 50% premium rank + 50% demand rank, both among the finalists	70/30, 30/70; ranks vs. all counties	Our ideas weigh price advantage and demand equally; same scale so weights are real
D6	No cap per metro area	Max. one or two per metro	Neighboring counties are separate local markets; we want the best counties
D7	Unweighted cost model	Weighted by children (WLS)	Every county is one market; size enters through D4 and D5
D8	Plan with the expected premium (after typical fading)	Use today's premium	Premiums fade partially (D-stability below)

D2. The rule, step by step

STEP	COUNTIES	CRITERION	DATA
Metro counties with price, wage and rent	1,159	In the analysis dataset at least once	NDCP × OEWS × ACS, 2015–2022
Eligible	972	In the 2022 data and observed in ≥ 5 of 8 years	Current price + enough history
Positive nationally	460	Average national premium > 0	Same model without state effects
Candidate pool	20	Largest average within-state premium	Mean of yearly premiums
Persistence	20	Premium > 0 in every observed year	Yearly premiums
Market size	10	≥ 4,286 children needing care (2022 median)	ACS 2022
Top 10	10	Ordered by the score (removes none)	50% premium + 50% demand rank

D3. The candidate pool

#	COUNTY	AVG. PREMIUM	AVG. NATIONAL	WEAKEST YEAR	YEARS > 0	CHILDREN NEEDING CARE	PERSIST.	MARKET	FINAL RANK
1	Arlington County, VA	+61%	+46%	+47%	7/7	11,577	✓	✓	**1**
2	Alexandria City, VA	+50%	+33%	+35%	7/7	8,813	✓	✓	**2**
3	Manassas Park City, VA	+36%	+21%	+30%	7/7	816	✓	×	–
4	San Francisco County, CA	+36%	+21%	+21%	7/7	30,027	✓	✓	**3**
5	Loudoun County, VA	+35%	+21%	+31%	7/7	22,329	✓	✓	**4**
6	Falls Church City, VA	+33%	+19%	+28%	7/7	578	✓	×	–
7	Olmsted County, MN	+33%	+65%	+32%	8/8	9,533	✓	✓	**5**
8	Hall County, GA	+33%	+21%	+28%	8/8	9,215	✓	✓	**9**
9	Walker County, AL	+33%	+24%	+21%	8/8	2,173	✓	×	–
10	Fairfax City, VA	+32%	+18%	+26%	7/7	1,685	✓	×	–
11	Belmont County, OH	+32%	+64%	+17%	8/8	2,028	✓	×	–
12	Multnomah County, OR	+32%	+32%	+22%	8/8	32,227	✓	✓	**7**
13	Howard County, MD	+32%	+32%	+25%	8/8	16,287	✓	✓	**8**
14	Davie County, NC	+31%	+15%	+1%	7/7	1,845	✓	×	–
15	Colonial Heights City, VA	+31%	+35%	+28%	7/7	966	✓	×	–
16	Fairfax County, VA	+30%	+17%	+26%	7/7	57,257	✓	✓	**6**
17	Charlottesville City, VA	+30%	+23%	+23%	7/7	1,711	✓	×	–
18	Blount County, AL	+29%	+21%	+18%	8/8	2,477	✓	×	–
19	Jefferson County, OH	+28%	+57%	+20%	8/8	2,391	✓	×	–
20	Sullivan County, TN	+28%	+25%	+24%	8/8	5,603	✓	✓	**10**

All 20 candidates pass the persistence test; the ten that drop out fail only the market-size test (e.g. Falls Church City, VA, and Manassas Park City, VA, are too small). Not eligible because NDCP has no 2022 price for their state: St. Louis County, MO (average premium +29.5%, last price 2020); Washington County, PA (average premium +28.5%, last price 2020); both would otherwise have entered the pool and are worth a look once current prices are available.

D4. The score

Premium strength = percentile rank of the average premium among the ten; demand index = mean of the percentile ranks of children needing care, care-need share and income per capita among the ten; score = 0.5 × premium strength + 0.5 × demand index. Because both parts are ranked on the same scale, the 50/50 weights are the effective weights (an earlier version ranked demand against all counties, which tilted the weights; corrected). The score only orders the ten: 70/30 and 30/70 weights give the same ten counties.

RANK	COUNTY	AVG. PREMIUM	PREMIUM STRENGTH	CHILDREN	CARE-NEED SHARE	INCOME	DEMAND INDEX	SCORE
1	Arlington, VA	+61%	1.00	0.50	0.60	0.90	0.67	**0.83**
2	Alexandria, VA	+50%	0.90	0.20	0.90	0.80	0.63	**0.77**
3	San Francisco, CA	+36%	0.80	0.80	0.10	1.00	0.63	**0.72**
4	Loudoun, VA	+35%	0.70	0.70	0.80	0.60	0.70	**0.70**
5	Olmsted, MN	+33%	0.60	0.40	1.00	0.40	0.60	**0.60**
6	Fairfax, VA	+30%	0.20	1.00	0.70	0.70	0.80	**0.50**
7	Multnomah, OR	+32%	0.40	0.90	0.30	0.30	0.50	**0.45**
8	Howard, MD	+32%	0.30	0.60	0.50	0.50	0.53	**0.42**
9	Hall, GA	+33%	0.50	0.30	0.40	0.20	0.30	**0.40**
10	Sullivan, TN	+28%	0.10	0.10	0.20	0.10	0.13	**0.12**

D4b. Which market characteristics we use, and why

Our characteristics (assignment item c): the **average premium** and its **weakest year** (price advantage and its persistence), and three demand measures from the 2022 ACS: **children needing care** (under 6, all resident parents working: market size), their **share of the population** (need intensity) and **income per capita** (ability to pay). Two further ACS variables from Problem Set 5 are not used: the female college share correlates 0.83 with income (it would count affluence twice), and the female full-time share overlaps the care-need share, which measures working parents directly (0.36).

Within states, the demand measures are only weakly related to the premium (regression below): demand adds information the premium does not already contain, which is why the score weighs both. We deliberately did not pick demand variables by how well they predict the premium; that would count the premium twice.

AVERAGE LOG PREMIUM ON (STANDARDIZED, STATE EFFECTS)	USED IN THE SCORE	+ TWO UNUSED SHARES
Children needing care (log)	+1.6%* (0.007)	+0.5% (0.008)
Care-need share	−0.6% (0.008)	0.0% (0.008)
Income per capita (log)	+2.0%* (0.009)	−0.3% (0.009)
Female college share	–	+3.6%** (0.011)
Female full-time share	–	−0.7% (0.005)
R^2 (with state effects) / N	0.051	0.084

Coefficient = % change in the average premium per standard deviation; SE clustered by metro area; N = 972 counties (in the 2022 data, observed ≥ 5 years). ** p < 0.01, * p < 0.05.

D5. Are premiums stable over time?

OUR IDEA

A price advantage is only worth building on if it lasts. We test whether counties that had a premium in the past still have one later, and how much of it survives.

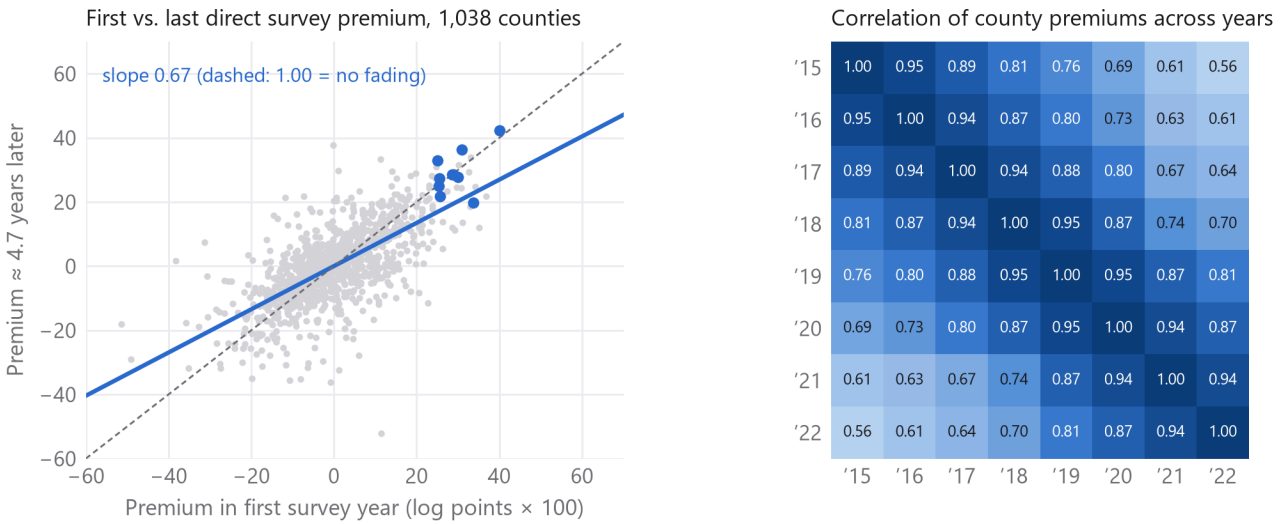


Figure D1. Left: within-state premium in a county's first and last direct NDCP survey year (≈ 4.7 years apart; blue = top 10). Right: correlation of county premiums between years, all counties.

Why survey years matter. NDCP interpolates prices between survey years, which mechanically smooths premiums and would overstate their stability. The cleanest test therefore uses only years in which both prices are direct survey values (5th digit of the NDCP imputation code = 0): one pair per county, the first and last survey year, 1,038 counties in 46 states, on average 4.7 years apart.

0.67
of a premium survives ≈ 4.7 years later (95% CI 0.57–0.78, SE clustered by state)

0.67
correlation of the two survey premiums

74%
of counties keep the sign of their premium

< 1
premiums fade partially (regression to the mean, plus noise in the first survey price)

Balanced panel (counties observed in all 8 years): regression of the 2022 premium on an earlier year's premium (log points, SE clustered by state).

MODEL	N	SLOPE	SE	R^2
2022 on 2015	712	0.59	0.07	0.31
2022 on 2018	712	0.73	0.07	0.44
2022 on 2019	712	0.91	0.05	0.64
2022 on 2021	712	1.01	0.02	0.89

Quintile transitions (balanced panel): counties in the top fifth of premiums in an earlier year, and where they are in 2022.

TOP FIFTH IN	N	TOP FIFTH 2022	ABOVE MEDIAN	POSITIVE
2015	143	56%	81%	82%
2018	143	63%	88%	89%
2019	143	73%	96%	97%
If premiums were random	–	20%	50%	≈ 50%

TEAM DECISION D8 · PLAN WITH THE EXPECTED PREMIUM

Because premiums fade partially, we do not promise today's premium. Expected premium = exp(a + b · average log premium) − 1, with a and b from the survey-only regression (b = 0.674); the range uses the 95% CI of b, so it reflects the uncertainty of the fading slope, not how much one county scatters around it. The expected premium is computed on the log scale, so the "average" used here is the geometric mean (column 4), which differs slightly from the arithmetic average used for ranking (column 3).

RANK	COUNTY	AVERAGE PREMIUM (ARITHMETIC)	AVERAGE (GEOMETRIC)	EXPECTED PREMIUM	RANGE (SLOPE CI)
1	Arlington, VA	+61%	+60%	**+37%**	[+31%, +44%]
2	Alexandria, VA	+50%	+49%	**+31%**	[+26%, +37%]
3	San Francisco, CA	+36%	+35%	**+23%**	[+19%, +27%]
4	Loudoun, VA	+35%	+35%	**+22%**	[+19%, +26%]
5	Olmsted, MN	+33%	+33%	**+21%**	[+18%, +25%]
6	Fairfax, VA	+30%	+30%	**+19%**	[+16%, +23%]
7	Multnomah, OR	+32%	+32%	**+20%**	[+17%, +24%]
8	Howard, MD	+32%	+31%	**+20%**	[+17%, +24%]
9	Hall, GA	+33%	+33%	**+21%**	[+18%, +25%]
10	Sullivan, TN	+28%	+28%	**+18%**	[+15%, +21%]

APPENDIX E · SLIDE 4

How reliable is the prediction?

OUR IDEA

Test the whole approach the way it would have been used: decide with old data only, then check what happened. And make sure the list does not hinge on any single design choice.

E1. Backtest: the complete rule as if we had decided in 2018

Every step of the rule (premiums, average over 2015–2018 with at least 3 years, national premium, persistence, market size and demand from 2018, score) applied with data up to 2018 only; outcome = the county's within-state premium in 2022, four years later.

PICKED IN 2018	AVG. 2015–18	PREMIUM 2022	PCT. 2022	TOP 10 TODAY
Arlington County, VA	+50%	+102%	100th	✓
Alexandria City, VA	+39%	+84%	100th	✓
Olmsted County, MN	+34%	+33%	98th	✓
Multnomah County, OR	+36%	+22%	94th	✓
St. Louis County, MO	+33%	no 2022 price	–	–
Loudoun County, VA	+33%	+38%	99th	✓
Hall County, GA	+33%	+30%	97th	✓
Milwaukee County, WI	+30%	+15%	86th	–
Washington County, PA	+31%	no 2022 price	–	–
Sullivan County, TN	+28%	+25%	96th	✓

+43.5%
average 2022 premium of the 8 picks still in the 2022 data

+0.6%
all other 990 counties

+25.0%
the same without CA/VA (5 picks), whose 2022 prices are extrapolated

+0.6%
all other counties without CA/VA (877)

100%
of the picks still have a positive premium (others: 49%)

7/10
of the 2018 picks are in today's top 10

Two honest caveats. The headline +43.5% includes Arlington and Alexandria, whose 2022 prices are NDCP extrapolations; without CA/VA the picks still beat all other counties by about 24 percentage points. And today's ranking averages 2015–2022, which includes the 2015–2018 years the backtest used, so the overlap of 7 of 10 is partly mechanical; the 2022 outcome itself is not.

E2. Robustness: the whole rule rerun with one change at a time

VARIANT	SAME	DROPPED	ADDED	STATUS
Previous design: pool ranked on 2022 premium only	6/10	Fairfax County, VA; Hall County, GA; Multnomah County, OR; Sullivan County, TN	Alameda County, CA; Marin County, CA; McLean County, IL; San Mateo County, CA	not counted: replaced by D2 (2022 extrapolations)
Average premium 2015–2020 (excludes NDCP 2022 extrapolation)	9/10	Sullivan County, TN	Milwaukee County, WI	counted
Average premium 2015–2019 (pre-COVID)	9/10	Howard County, MD	Milwaukee County, WI	counted
Level instead of log premium (LS, $/week, state effects)	7/10	Hall County, GA; Howard County, MD; Sullivan County, TN	Milwaukee County, WI; Norfolk County, MA; Ramsey County, MN	counted

VARIANT	SAME	DROPPED	ADDED	STATUS
National premium, no state effects (LL)	1/10	Alexandria City, VA; Arlington County, VA; Fairfax County, VA; Hall County, GA; Howard County, MD; Loudoun County, VA; Multnomah County, OR; San Francisco County, CA; Sullivan County, TN	Brown County, WI; Dane County, WI; Milwaukee County, WI; Outagamie County, WI; Racine County, WI; Ramsey County, MN; Sheboygan County, WI; Waukesha County, WI; Winnebago County, WI	not counted: rejected in D1 (selects states, not markets)
Weights 70% premium / 30% demand	10/10	–	–	not counted: cannot change the list by construction
Weights 30% premium / 70% demand	10/10	–	–	not counted: cannot change the list by construction
Persistence: one negative year allowed	10/10	–	–	not counted: cannot change the list by construction
Market threshold: 25th percentile	9/10	Sullivan County, TN	Fairfax City, VA	counted
Candidate pool: top 30 instead of top 20	9/10	Sullivan County, TN	Ramsey County, MN	counted

Counted variants: 5 (the two weight variants and "one negative year allowed" are reported but cannot change the list by construction: the score only orders the ten, and all 20 candidates already pass the persistence filter). Selected in all counted variants: 7 of 10; Howard, MD 3/5; Hall, GA 4/5; Sullivan, TN 1/5.

E2b. Portfolio view: at most two counties per metro area

Not a robustness test but a business view of the Northern Virginia cluster (Appendix F3): the same rule, walking down the ranking until ten counties qualify with at most two per metro area. The two lower-ranked Northern Virginia counties give way to the next qualifying markets; the other eight stay.

RANK	COUNTY	METRO AREA	AVG. PREMIUM	SCORE	IN TOP 10
1	Arlington County, VA	Washington-Arlington-Alexandria	+61%	0.87	✓
2	Alexandria City, VA	Washington-Arlington-Alexandria	+50%	0.78	✓
3	San Francisco County, CA	San Francisco-Oakland-Hayward	+36%	0.72	✓
4	Olmsted County, MN	Rochester	+33%	0.70	✓
5	Multnomah County, OR	Portland-Vancouver-Hillsboro	+32%	0.57	✓
6	Howard County, MD	Baltimore-Columbia-Towson	+32%	0.53	✓
7	Hall County, GA	Gainesville	+33%	0.50	✓
8	Ramsey County, MN	Minneapolis-St. Paul-Bloomington	+27%	0.40	**new**
9	Sullivan County, TN	Kingsport-Bristol-Bristol	+28%	0.23	✓
10	Trumbull County, OH	Youngstown-Warren-Boardman	+28%	0.20	**new**

E3. Different wage measures and weighting

SPECIFICATION	SAME	DROPPED	ADDED
Baseline: childcare-worker wage (SOC 39-9011)	10/10	–	–
Preschool-teacher wage (SOC 25-2011)	8/10	Hall County, GA; Sullivan County, TN	Clayton County, GA; Ramsey County, MN
Both wages	10/10	–	–

SPECIFICATION	SAME	DROPPED	ADDED
Weighted by children under 6 (WLS)	6/10	Fairfax County, VA; Loudoun County, VA; Multnomah County, OR; Olmsted County, MN	Erie County, NY; McLean County, IL; Onondaga County, NY; St. Johns County, FL

Centers employ preschool teachers (SOC 25-2011) as well as childcare workers (SOC 39-9011, the assignment's measure). With both wages in the 2022 model, the childcare-worker elasticity is 0.33 vs. 0.06 for teachers, and R^2 barely changes (sample with both wages, N = 938) (0.811 vs. 0.812). Weighting by children lets large counties dominate the cost line; we keep every county as one market observation (D7).

E4. Statistical checks

Average premium (what we select on): z-score and percentile among all 1,059 counties observed in ≥ 5 years (log scale). **Single year 2022:** studentized residual of the final model and whether the county lies above the 95% prediction band.

RANK	COUNTY	Z (AVERAGE)	PERCENTILE (AVERAGE)	PREMIUM 2022	STUDENTIZED 2022	ABOVE 95% BAND 2022
1	Arlington, VA	4.36	100.0	+102%	5.21	✓
2	Alexandria, VA	3.70	99.9	+84%	4.48	✓
3	San Francisco, CA	2.81	99.7	+66%	3.73	✓
4	Loudoun, VA	2.76	99.6	+38%	2.33	✓
5	Olmsted, MN	2.66	99.5	+33%	2.08	✓
6	Fairfax, VA	2.42	98.7	+27%	1.76	–
7	Multnomah, OR	2.53	99.0	+22%	1.48	–
8	Howard, MD	2.53	98.9	+36%	2.28	✓
9	Hall, GA	2.63	99.3	+30%	1.91	–
10	Sullivan, TN	2.30	98.0	+25%	1.64	–

All ten average premiums are in the top 2% (z 2.3–4.4). This describes how extreme they are; it is not an independent test, because the candidate pool is the 20 largest averages. In 2022 alone, 6 of the ten lie above the 95% prediction band, against 2.4% of all counties; four of those six have extrapolated CA/VA 2022 prices. The other four have clear but not extreme 2022 premiums and qualify through their persistence over all years: a single year is noisy, a premium that holds for 7–8 years is not.

E5. Data quality: are the premiums real survey prices?

RANK	COUNTY	SURVEY YEARS	WHICH	AVG. PREMIUM, SURVEY YEARS	AVG. PREMIUM, ALL YEARS
1	Arlington, VA	2 of 7	2015, 2018	+51%	+61%
2	Alexandria, VA	2 of 7	2015, 2018	+40%	+50%
3	San Francisco, CA	2 of 7	2016, 2018	+30%	+36%
4	Loudoun, VA	2 of 7	2015, 2018	+34%	+35%
5	Olmsted, MN	3 of 8	2016, 2018, 2021	+33%	+33%
6	Fairfax, VA	2 of 7	2015, 2018	+29%	+30%
7	Multnomah, OR	4 of 8	2016, 2018, 2020, 2022	+31%	+32%
8	Howard, MD	2 of 8	2017, 2021	+34%	+32%
9	Hall, GA	2 of 8	2016, 2021	+33%	+33%
10	Sullivan, TN	5 of 8	2015, 2016, 2017, 2018, 2021	+28%	+28%

NDCP surveys each state every few years and interpolates (or, for CA and VA in 2022, extrapolates) in between. Every top-10 county also has a large premium in its direct survey years, so the advantage is not an artifact of interpolation.

APPENDIX F · SLIDE 6

What the model does not see yet

OUR IDEA

A premium tells us where parents pay more than costs justify, not whether a new center would succeed there. Before recommending a market we check what could erode the premium: competition, shrinking demand, rising costs and free public alternatives.

F1. Risk checks and thresholds

CHECK	MEASURE	FLAG (VS. QUARTILES OF ALL METRO COUNTIES WITH DATA)	COUNTIES
Competition	CBP child day care establishments per 1,000 children needing care (2022)	! above 6.24 (crowded) or below 3.86 (untested market)	882
Child trend	Change in children under 6, ACS 2015 → 2022	! below median (−4.4%); ✕ bottom quarter (below −10.0%)	903
Wage growth	Real childcare-worker wage, 2019 → 2022	! top quarter (above +6.1%)	974
Rent growth	Real median gross rent, 2015 → 2022	! top quarter (above +12.0%)	903
Public pre-K	Free public pre-K for all incomes (universal) in the county	! universal program	10 (sourced)

COUNTY	CENTERS 2022	PER 1,000 CHILDREN	CHILDREN <6, 2015–22	WAGE, 2019–22	RENT, 2015–22
Arlington, VA	63	5.44	−3.3%	+5.0%	−1.3%
Alexandria, VA	50	5.67	−5.9%	+5.0%	+3.3%
San Francisco, CA	427	14.22	−5.5%	−1.3%	+20.4%
Loudoun, VA	146	6.54	+0.8%	+5.0%	+6.6%
Olmsted, MN	50	5.24	−5.7%	−1.9%	+16.8%
Fairfax, VA	356	6.22	−9.5%	+5.0%	+0.1%
Multnomah, OR	392	12.16	−15.8%	+7.6%	+27.3%
Howard, MD	114	7.00	+0.5%	+8.6%	−1.5%
Hall, GA	24	2.60	−5.1%	+13.3%	+11.3%
Sullivan, TN	17	3.03	−6.4%	+6.4%	+6.7%

CBP counts employer establishments (not capacity, quality, or home-based providers without employees). Across all counties, premium and center density are only weakly related, so a premium is not simply a sign of missing supply. **Recommendation rule:** start where at most one minor flag appears (Arlington, Alexandria, Loudoun, Fairfax: one Northern Virginia market, F3); next in line are Olmsted and Howard.

F2. Free public pre-K (checked against program rules)

COUNTY	PROGRAM	SCOPE	ELIGIBILITY	FLAG	SOURCE
Arlington, VA	Virginia Preschool Initiative (VPI) / Mixed Delivery	targeted	family income at or below 200% of federal poverty guidelines or other at-risk criteria	–	https://nieer.org/sites/default/files/2023-08/virginia_yb2022.pdf
Alexandria, VA	Virginia Preschool Initiative (VPI) / Mixed Delivery	targeted	family income at or below 200% of federal poverty guidelines or other at-risk criteria	–	https://nieer.org/sites/default/files/2023-08/virginia_yb2022.pdf
Loudoun, VA	Virginia Preschool Initiative (VPI) / Mixed Delivery	targeted	family income at or below 200% of federal poverty guidelines or other at-risk criteria	–	https://nieer.org/sites/default/files/2023-08/virginia_yb2022.pdf
Fairfax, VA	Virginia Preschool Initiative (VPI) / Mixed Delivery	targeted	family income at or below 200% of federal poverty guidelines or other at-risk criteria	–	https://nieer.org/sites/default/files/2023-08/virginia_yb2022.pdf

COUNTY	PROGRAM	SCOPE	ELIGIBILITY	FLAG	SOURCE
San Francisco, CA	California universal transitional kindergarten (TK) + San Francisco Preschool for All	universal	TK: free for every child turning 4 by Sept 1 (full implementation 2025-26); city Preschool for All: all San Francisco 4-year-olds regardless of income	!	https://www.cde.ca.gov/NR/ne/yr23/yr23rel35.asp
Olmsted, MN	Minnesota Voluntary Prekindergarten (VPK)	targeted	free for 4-year-olds meeting at least one at-risk criterion; limited state-funded seats; others fee-based	–	https://dcyf.mn.gov/partners-and-providers/child-care-and-early-learning/districts-schools-and-head-start/preschool
Multnomah, OR	Multnomah County Preschool for All	universal (phasing in)	free for 3- and 4-year-olds of county residents; started 2022-23; scaling to universal access by 2030	!	https://multco.us/preschool/families
Howard, MD	Maryland Blueprint prekindergarten	targeted (income-tiered)	free full-day pre-K at or below 300% of federal poverty level; sliding-scale cost for 4-year-olds at 300-600%	–	https://blueprint.marylandpublicschools.org/?p=2052
Hall, GA	Georgia's Pre-K Program (lottery-funded)	universal	every Georgia resident who is 4 on September 1; no income test	!	https://www.decal.ga.gov/PreK/About.aspx
Sullivan, TN	Tennessee Voluntary Pre-K (VPK)	targeted	economically disadvantaged (185% of poverty guidelines) and other at-risk children; limited classrooms	–	https://tennessee.gov/education/districts/early-learning/voluntary-pre-k/voluntary-pre-k-information-for-parents.html

Universal programs compete directly with paid preschool for 4-year-olds (Georgia's lottery-funded Pre-K; California's universal transitional kindergarten, fully phased in for 2025–26, plus San Francisco's Preschool for All; Multnomah's Preschool for All, ages 3–4, scaling to universal access by 2030). Income-targeted programs (VA, MD, MN, TN) mainly serve families outside the target segment of a premium-priced center. Georgia's program dates from the 1990s, so it is already reflected in Hall's prices; California's TK and Multnomah's Preschool for All started during or after our price data and are a forward-looking risk. Sources accessed in October 2026 (`data/manual/public_prek_programs.csv`).

F3. Four of ten in one market: the Northern Virginia cluster

Arlington, Alexandria, Loudoun and Fairfax (ranks 1, 2, 4, 6) are neighbors in the Washington metro area. We checked whether this is a real market effect or an artifact of the data:

- **A sharp inner core, not the whole metro.** Of the 23 Washington-metro counties with enough history, the inner Northern Virginia jurisdictions all have premiums of about +30% to +61%, while outer counties are near zero or negative (e.g. Prince William +12%; Stafford −4%; Prince George's −5%; Fauquier −16%).
- **Real survey prices, national comparison.** The four keep large premiums in their direct survey years (E5) and against the national model (Arlington +46%); all four stay selected without the 2022 data (E2).
- **Partly structural.** Virginia reports many small jurisdictions (80 in our data, incl. independent cities), and its premiums are the 3rd most dispersed of 34 states (SD 0.160 vs. median 0.116). More, smaller units produce more extreme values: 9 of the 20 candidates are Virginian; the market-size filter keeps only the four large ones.
- **Not four independent confirmations.** They share one OEWS wage, one labor market, one state survey (direct prices only in 2015 and 2018) and the same exposure to federal employment changes. We therefore recommend one first site in Northern Virginia, with current prices checked on the ground, and show a portfolio view with at most two counties per metro area (E2b).

F4. Other promising counties (teal on the slide 1 map; St. Louis and Trumbull not shown)

COUNTY	EVIDENCE	WHY NOT IN THE TOP 10
Washington, DC	+71% in 2022 when grouped with MD and VA	NDCP price only for 2022: no history to test persistence; with state effects a one-county "state" has a premium of 0 by construction
Milwaukee, WI	Selected when 2022 is excluded, pre-COVID only, or with $ premiums	Robustness variants (E2)
Ramsey (St. Paul), MN	Selected with $ premiums, a top-30 pool, the preschool-teacher wage, and in the portfolio view	Robustness variants (E2, E2b, E3)

COUNTY	EVIDENCE	WHY NOT IN THE TOP 10
Norfolk, MA	Selected with $ premiums	Robustness variant (E2)
Fairfax City, VA	Selected with the smaller market threshold	Robustness variant (E2)
Clayton, GA	Selected with the preschool-teacher wage	Robustness variant (E3)
St. Louis, MO	Average premium +29.5%; picked by the 2018 backtest	No NDCP price after 2020 (D3)
Trumbull (Youngstown), OH	Enters the portfolio view with at most 2 counties per metro area (E2b)	Lower premium and demand than the ten

APPENDIX G

Limitations and reproducibility

G1. Limitations

- **Prices are survey medians, not offers.** NDCP prices are county medians from state market-rate surveys; each top-10 county has only 2–5 direct survey years. A new premium provider may be able to charge more or less than the median.
- **Costs are proxies.** We use residential rent (a center pays commercial rent) and average metro wages (not vacancies or turnover); all counties of a metro area share one wage.
- **The residual mixes signal and noise.** Part of each premium reflects quality, regulation or measurement error. Averaging over years, the persistence filter, the survey-only checks and the robustness variants reduce but cannot remove this.
- **Demand is measured once.** Market size and the demand score use the 2022 ACS; post-COVID shifts such as remote work are not captured.
- **Associations, not causal effects.** The cost model describes how prices co-move with costs; it is used to measure unusual prices, not to claim what would happen if wages changed.
- **Free public pre-K** is universal in three of the ten counties (new or expanding in San Francisco and Multnomah); its future effect on paid preschool demand is not in our price data.

G2. Reproducibility

`python run_pipeline.py` reruns all scripts in order from the included data (no downloads or keys needed; see the README; `--full` re-downloads the raw data). Scripts marked "homework only" are not needed for the project (Python 3.14; pandas, statsmodels, scikit-learn, scipy, matplotlib, python-pptx, pyshp; versions in `requirements.txt`). The raw BLS OEWS zip files (275 MB) are not included; `02_get_oews.py` downloads them, and the extracted wages are in `data/interim/`. Downloaded raw files are never edited; 19 files are fixed by SHA-256 checksums in `data/raw_checksums.sha256`. The Census API key is read from a local, git-ignored file (`census_api_key.txt`) and is not part of the submission; BLS downloads use the contact e-mail of whoever runs the code (environment variable `BLS_CONTACT_EMAIL`). **Number check:** `35_fp_check_deck_numbers.py` recomputes every number stated on the slides from the result tables (83 of 83 claims PASS, 0 FAIL).

SCRIPT	ROLE	PURPOSE
`01_get_cpi.py`	data	Download CPI-U (BLS series CUUR0000SA0) and build annual averages + 2022-dollar deflators.
`02_get_oews.py`	data	Download BLS OEWS metro/nonmetro area files (2015-2022) and extract SOC 39-9011 Childcare Workers.
`03_get_acs.py`	data	Pull ACS 5-year county estimates (2015-2022) from the Census API for the project variables.
`04_prepare_ndcp.py`	data	Extract the NDCP childcare price series we need (2015-2022) into a tidy county x year file.
`05_build_metro_sample.py`	data	PS4 Q1: restrict NDCP county prices to the metro analysis sample.
`06_ps4_q2_distribution_2022.py`	homework only	PS4 Q2: distribution of 2022 median weekly center-based prices by child age group (metro sample).
`07_ps4_q2_factsheet.py`	homework only	PS4 Q2 supplementary descriptives for the written discussion (2022, metro sample).
`08_ps4_q3_real_trends.py`	data	PS4 Q3: convert NDCP prices to constant 2022 dollars (CPI-U) and plot trends by age group.
`09_ps4_q4_voucher_test.py`	homework only	PS4 Q4: test whether the CCDF monthly voucher equals the mean price of median-priced center-based preschool care across metro counties.
`10_ps4_q4_robustness.py`	homework only	PS4 Q4 robustness: does the voucher test depend on the unit of observation / independence assumption?
`11_build_submission.py`	homework only	Build the Problem Set 4 submission PDF (MGT 403, Homework 4, Question 2).
`12_build_acs_derived.py`	data	PS5 task 1: construct the derived ACS variables defined in the assignment (county x year, 2015-2022).

SCRIPT	ROLE	PURPOSE
`13_build_cost_panel.py`	data	PS5 task 3: merge OEWS area wages down to counties, attach ACS county data, keep metro counties.
`14_ps5_q4_descriptives_2022.py`	homework only	PS5 task 4: descriptive table (mean / SD / N) of each predictor, metro counties, 2022.
`15_ps5_q5_real_trends.py`	data	PS5 task 5: convert dollar values to constant 2022 dollars (CPI-U) and plot trends in childcare-worker wages and rents.
`16_ps5_q6_wage_test.py`	homework only	PS5 task 6: were average inflation-adjusted childcare-worker wages the same in 2022 as in 2019?
`17_ps5_q7_regression.py`	homework only	PS5 task 7: bivariate regression (2022) of childcare-worker wages on female college share.
`18_pre_merge_prices_costs.py`	pre-analysis	Pre-analysis: merge NDCP county prices onto the year x county cost/demand file from Problem Set 5.
`19_pre_analysis_dataset.py`	pre-analysis	Pre-analysis task 1: define the analysis dataset = year x county observations where the price and all predictor variables are non-missing (complete cases).
`20_pre_t2_descriptives_2022.py`	pre-analysis	Pre-analysis task 2: mean / SD / N for all variables in the analysis dataset, 2022.
`21_pre_t3_scatter.py`	pre-analysis	Pre-analysis task 3: scatter plots of the preschool price against (a) childcare-worker wages and (b) rents, 2022 analysis dataset, each with the fitted line from a bivariate OLS regression.
`22_pre_t4_regressions.py`	pre-analysis	Pre-analysis task 4: 2022 regressions of the median preschool price on (a) wages, (b) rent, (c) both.
`23_pre_t4_coef_plot.py`	pre-analysis	Pre-analysis task 4 figure: wage and rent coefficients alone vs. together (2022, metro-clustered 95% CIs).
`25_fp_premium_models.py`	final project	Final project step 1: cost-based price models, historical validation, and the price premium.
`26_fp_persistence_demand.py`	final project	Final project step 2: persistence and demand facts for the top-20 candidates, and a historical test of whether premiums persist.
`27_fp_select_top10.py`	final project	Final project step 3: select the top 10 counties (team decisions) and test the robustness of the selection.
`28_fp_math_checks.py`	final project	Final project step 4: methodological checks of the selection.
`29_fp_stability.py`	final project	Final project (d): are prices that exceed cost-based predictions stable over time?
`30_fp_missing_factors.py`	final project	Final project (e): factors the prediction does not include, quantified where public data allow.
`31_fp_improvements.py`	final project	Final project: four improvements / robustness checks of the selection.
`32_fp_analysis_register.py`	final project	Analysis register: for every analysis in the final project, the sample, variables, method, N and output file.
`33_fp_figures.py`	final project	Presentation figures (main deck), rendered at their EXACT slot size on the slide grid (no automatic cropping), so that plot edges align with the slide grid and figure type matches slide type 1:1 (9-10.5 pt, min. 7.5 pt).
`34_build_deck.py`	final project	Build the final-project slide deck (PowerPoint, 16:9) in a clean, Apple-inspired style.
`35_fp_check_deck_numbers.py`	final project	Number check: every number stated on the slides is recomputed from the result tables / analysis data and searched for in the text of the built deck (project/deck/MGT403_FinalProject_Deck.pptx).
`36_fp_appendix_analysis.py`	final project	Appendix: supporting tables and figures (recomputed from the analysis data; nothing here changes the selection).
`37_build_appendix.py`	final project	Build the combined Pre-Analysis and Appendix document (HTML -> PDF with headless Microsoft Edge), in the design language of the deck.
`38_build_prompt_log.py`	final project	Build the prompt logs in the submitted format: the team's prompts only, verbatim (typos included), no AI output.

AI use: the analysis code, figures and drafts of this text were produced with Claude Code at the team's direction (https://claude.ai/code); every prompt, verbatim and dated, is in the prompt log submitted with the code (`MGT403_Prompt_Log.pdf`). Decisions marked as team decisions were taken by the team.